In [2]:
```
import pandas as pd
dataset=pd.read_csv("social_Network.csv")
dataset
```

Out[2]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

In [3]:
```
dataset=pd.get_dummies(dataset,drop_first=True)
dataset
```

Out[3]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	True
1	15810944	35	20000	0	True
2	15668575	26	43000	0	False
3	15603246	27	57000	0	False
4	15804002	19	76000	0	True
...	...	...	...	...	...
395	15691863	46	41000	1	False
396	15706071	51	23000	1	True
397	15654296	50	20000	1	False
398	15755018	36	33000	0	True
399	15594041	49	36000	1	False

400 rows × 5 columns

In [4]:
```
dataset=dataset.drop(columns=["User ID"])
dataset
```

Out[4]:

	Age	EstimatedSalary	Purchased	Gender_Male
0	19	19000	0	True
1	35	20000	0	True
2	26	43000	0	False
3	27	57000	0	False
4	19	76000	0	True
...	...	...	...	...
395	46	41000	1	False
396	51	23000	1	True
397	50	20000	1	False
398	36	33000	0	True
399	49	36000	1	False

400 rows × 4 columns

In [5]:
```
x=dataset[["Age","EstimatedSalary","Gender_Male"]]
y=dataset[["Purchased"]]
print(y)
```

```
     Purchased
0            0
1            0
2            0
3            0
4            0
..         ...
395          1
396          1
397          1
398          0
399          1

[400 rows x 1 columns]
```

In [7]:
```
dataset["Purchased"].value_counts()
```

Out[7]:

```
Purchased
0    257
1    143
Name: count, dtype: int64
```

In [6]:
```
dataset.shape
```

Out[6]:

```
(400, 4)
```

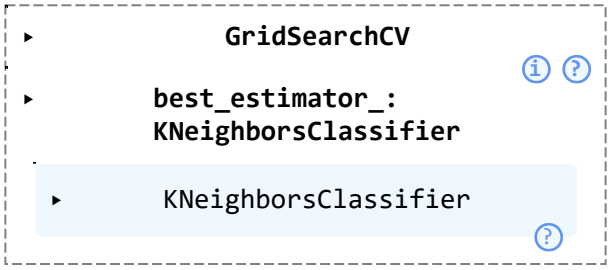
In [8]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=2/5,random_state=42)
```

In [30]:
```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import GridSearchCV
param_grid={
    'n_neighbors':[1,3,5],
    'weights':['uniform', 'distance'],
    'algorithm':['auto','ball_tree','kd_tree','brute']
}
clf=GridSearchCV(KNeighborsClassifier(),param_grid,refit=True,verbose=3,n_jobs=-1)
clf.fit(x_train,y_train)
```

```
Fitting 5 folds for each of 24 candidates, totalling 120 fits
```

```
C:\Users\User\anaconda3\Lib\site-packages\sklearn\neighbors\_classification.py:243:
DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change
the shape of y to (n_samples,), for example using ravel().
  return self._fit(X, y)
```

Out[30]:



In [31]:
```
y_pred=clf.predict(x_test)
print(y_pred)
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_pred,y_test)
print(cm)
```

```
[1 1 0 1 1 0 1 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 1 0 0 1 0 0 1 1 0 0 1 0 1 0 0
 0 1 0 1 0 0 1 0 1 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 1 0 1 0 0 1 0 0 0
 1 0 1 0 0 0 0 1 1 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 1 0 1 0 0 0 0 1 1 1 1 0 0
 0 0 0 0 0 0 1 0 0 0 0 0 1 1 1 0 0 0 1 1 0 1 0 0 0 0 1 1 0 1 0 0 0 0 1 1 0
 0 0 0 0 0 0 0 0 1 0 1 0]
[[89 23]
 [11 37]]
```

In [32]:
```
from sklearn.metrics import classification_report
report=classification_report(y_pred,y_test)
print(report)
```

```
              precision    recall  f1-score   support

           0       0.89      0.79      0.84       112
           1       0.62      0.77      0.69        48

    accuracy                           0.79       160
   macro avg       0.75      0.78      0.76       160
weighted avg       0.81      0.79      0.79       160
```

In [ ]: